

Data Viz DC Meetup + Ballston BID



LEARNING LAB SERIES



BALLSTON

Code of Conduct



Diversity Statement

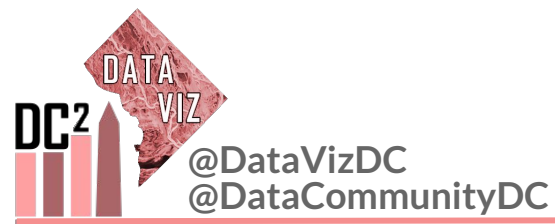
Data Community DC believes that diverse communities are better and stronger. Together, we continue to build an inclusive culture that encourages, supports, and celebrates diversity in race, gender, sexual orientation, religion, and all protected classes. It connects and brings us closer. Everyone is welcome to come, speak, and volunteer as an inclusive community.

Code of Conduct

www.dc2.org/code



DVDC Organizers



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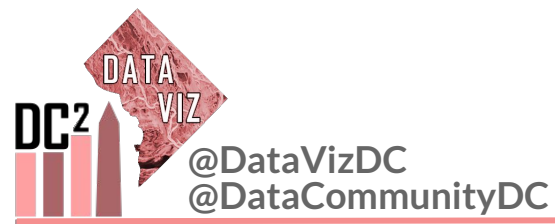


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Data Viz DC



Data Visualization DC is a non-profit professional group that meets monthly to discuss data visualization, data communication, infographics, exploratory data analysis, charts and graphs, perceptual psychology, and more! Our members are data journalists, data scientists, business intelligence professionals, and others who tell stories and persuade with data.



DC2 Sponsors

P R E F E C T  **ARLINGTON**
ECONOMIC DEVELOPMENT

Workshop: AI for data Viz

Agenda:

1. Brief overview of the fundamentals of data visualization
2. Methods of using AI for visualization
3. Pitfalls of AI for visualization
4. Hands on workshop with real data!

Brief overview of data viz:

Data visualization is the art and science of presenting information visually.

- Subfields include:
 - Business Intelligence
 - Information design
 - Cartography / GIS
(Geographical Information Systems)

Note: Data visualization is also a *skill* in addition to a job family.

The key considerations of data viz:

Data visualization is the art and science of **presenting** information visually.

Presenting is an act of **communication**

The best visualization starts with:

- Defining your objective
- Identify your audience and their needs.
- Determine the desired takeaway and intended actions.

Output formats

Data visualization artifacts (the created outputs) have two important dimensions:

1. Exploratory vs Explanatory
2. Static vs Dynamic

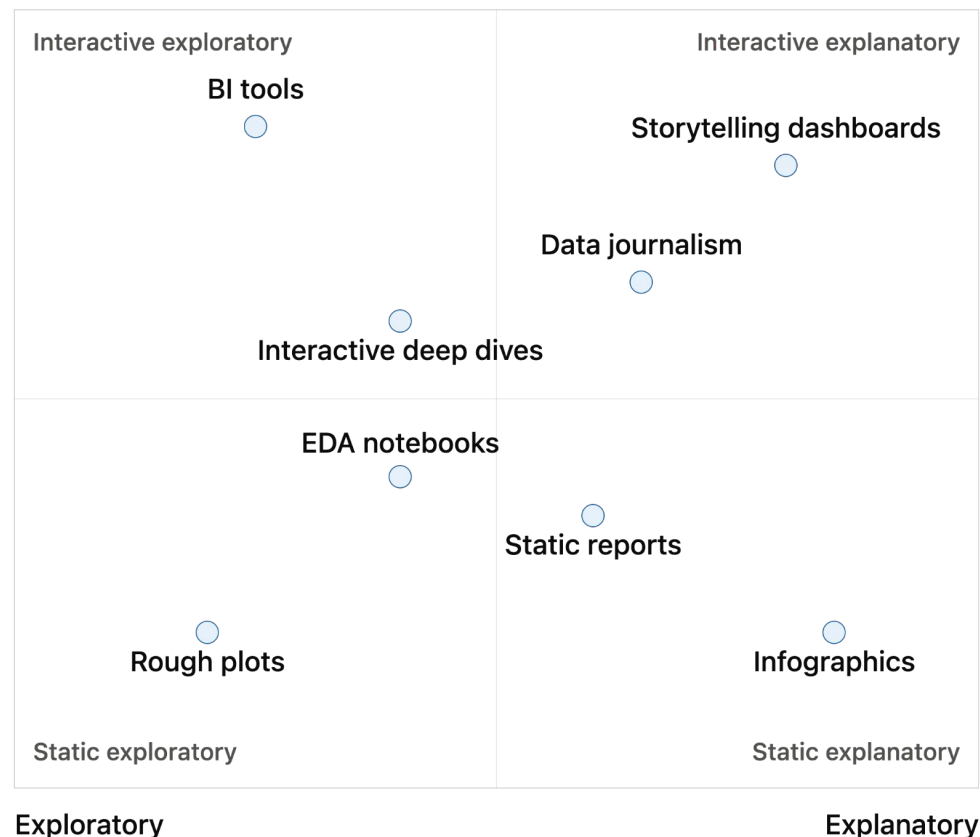
Mixing these two up will reduce the effectiveness of your communication.

Cole Nussbaumer Knaflig -

<https://www.storytellingwithdata.com/blog/2014/04/exploratory-vs-explanatory-analysis>

Interactive

Static



AI for data viz!

Core Benefit of AI: Rapid iteration and increased accessibility for non-experts.

AI approaches:

1. Direct Analysis: Using LLMs (ChatGPT, Claude, Gemini) for quick plots and infographics.
2. Code Generation: Using AI to write code for visualizations.
3. Tool Integration: Using AI to build interactive dashboards.

AI for data viz - notes

AI's Effectiveness for data visualization:

1. Straightforward Data:

AI is highly efficient and often correct (90-95% of the time).

2. Complex Data:

AI struggles; results become less reliable.

DANGEROUS

Pitfalls of AI for data viz:

AI IS NOT PERFECT

1. Accuracy: Risk of "hallucinations" or incorrect data analysis.
 - a. Some AI systems like to generate images which will not encode accurate information!
 - b. ACCURATE INFORMATION ENCODING DIRECTLY VIA DIFFUSION MODELS IS AN UNSOLVED PROBLEM**
 - i. Make sure the AI is giving you code.
 - ii. Make sure the AI is managing sources
2. Robustness: AI is less robust than a skilled human expert when handling non-obvious data.
3. Requirement: Human oversight is essential; users must remain suspicious and thoughtful regarding AI outputs.

Pitfalls: THE AI LIES



I'm glad you like it! One important caveat: the infographic image is not actually based on verified underlying data. The image model generated a plausible-looking design and invented a methodology and rankings. For example:

- "Instacart (2023) hot sauce sales per capita" — I don't know of a public state-by-state dataset like this.

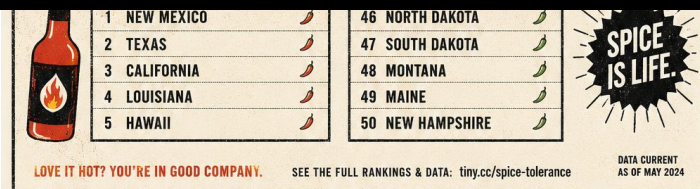
Keep the design simple and crisp with a hint of classic retro zine style graphic design.



I don't want to fabricate the numbers

I could make up plausible scores, but that defeats the purpose of a data-driven infographic.

Great. And the data sources?



<https://github.com/>

DataDrivenAngel/ai_for_data_viz

Workshop